

Robust Control Design for Linear Systems via Multiplicative Noise

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Model-based learning control methods

- System identification
- Adaptive control
- Reinforcement learning

⇒ Model estimates generated from data with quantifiable error

How can we design controllers robust to this model uncertainty?

- Relate **robust stability** and **stochastic stability**
- Guarantee robustness margins

True system

$$x_{t+1} = \bar{A}x_t + \bar{B}u_t$$



Nominal system

$$x_{t+1} = Ax_t + Bu_t$$

Robust stabilization problem

Find linear state-feedback $u_t = Kx_t$ such that

$$x_{t+1} = [(A + \Delta A) + (B + \Delta B)K]x_t$$

is stable for the largest set of $A + \Delta A$ and $B + \Delta B$.

Certainty-equivalent control

- Ignore the modeling error entirely
- High risk - no robustness to model errors

$\mathcal{H}_\infty$ robust control and dynamic game theory [1]

- Treat modeling error as an adversarial state-additive disturbance
- Excess conservatism

Shared Lyapunov functions via convex (semidefinite) programming [2, 3]

- Guarantee stability of any system in a polyhedral convex hull
- Choosing corner matrices is nontrivial

Dynamics:

$$x_{t+1} = \left(A + \sum_{i=1}^p \gamma_{ti} A_i \right) x_t + \left(B + \sum_{j=1}^q \delta_{tj} B_j \right) u_t$$

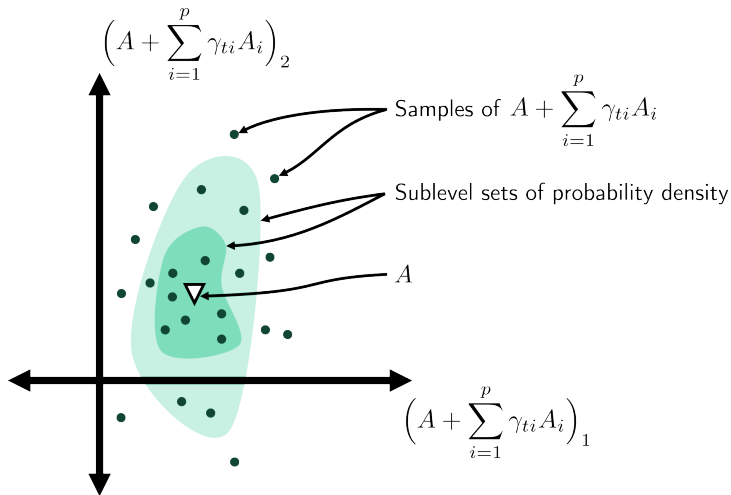
Multiplicative noise terms modeled by $\gamma_{ti} A_i$ and $\delta_{tj} B_j$

Scalars γ_{ti} , δ_{tj}

- Random variables
- i.i.d. across time (white)
- Mutually independent
- Zero-mean
- Variances α_i and β_j respectively

Matrices A_i and B_i

- Constants
- Specify how each scalar noise term affects the dynamics and input matrices



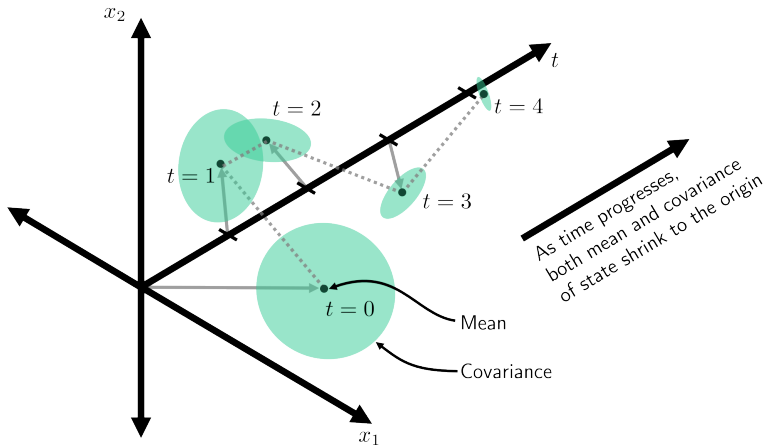
Mean-square stability [4, 2]

A stochastic system with $u_t = Kx_t$ is *mean-square stable* if and only if

$$\lim_{t \rightarrow \infty} \mathbb{E}[x_t x_t^\top] = 0 \quad \forall \quad \|x_0\| < \infty$$

if and only if for any $Q \succ 0$ there exists $P \succ 0$ satisfying

$$P = Q + (A + BK)^\top P (A + BK) + \sum_{i=1}^p \alpha_i A_i^\top P A_i + \sum_{j=1}^q \beta_j K^\top B_j^\top P B_j K.$$



Fundamental question

What is the set of ΔA , ΔB where the true system is guaranteed stable given knowledge only of A , B and mean-square stability of a system with multiplicative noise?

Fundamental idea

Restrict search to $\Delta A \in \sum_{i=1}^p \mu_i A_i$

Mean-square stability
under instantaneous
stochastic perturbations
in specific directions A_i
with variances α_i



Deterministic stability
under constant shifts of
the dynamics in those
same directions A_i with
magnitude $\mu_i \leq \sqrt{\alpha_i}$

Lemma 1 (Scalar case)

Suppose the stochastic system

$$x_{t+1} = (A + \gamma_t)x_t$$

*is mean-square stable where A , x_t , γ_t are scalars with $\mathbb{E}[\gamma_t^2] = \alpha > 0$.
Then, the perturbed deterministic system*

$$x_{t+1} = (A + y)x_t$$

is stable for any fixed perturbation $|y| \leq \sqrt{A^2 + \alpha} - |A| \leq \sqrt{\alpha}$.

Two important features:

- Margin increases monotonically with multiplicative noise variance
- When $\alpha = 0$ the bound collapses and no robustness is guaranteed

Theorem 2

Suppose $x_{t+1} = (A + \sum_{i=1}^p \gamma_{ti} A_i) x_t$ is mean-square stable with $\mathbb{E}[\gamma_{ti}] = 0$, $\mathbb{E}[\gamma_{ti}^2] = \alpha_i > 0$.

Fix a $Q \succ 0$ and the solution $P \succ 0$ to

$$P = pQ + A^\top P A + \sum_{i=1}^p \alpha_i A_i^\top P A_i.$$

Let $A = A_0$, $\eta_0 = 1$, and $\eta_i > 0$ be scalars which satisfy

$$pQ + \sum_{i=1}^p \alpha_i A_i^\top P A_i \succeq \sum_{i=0}^p \sum_{j=1}^p \eta_i \eta_j (A_i^\top P A_j + A_j^\top P A_i)^+.$$

Then $x_{t+1} = (A + \sum_{i=1}^p \mu_i A_i) x_t$ is deterministically stable for any $0 \leq \mu_i < \eta_i$.

Corollary 3

Make the assumptions of Theorem 2. Choose $\eta_i > 0$ proportional to $\sqrt{\zeta_i^2 + \alpha_i} - \zeta_i$, where ζ_i are scalars which marginally satisfy

$$\frac{1}{\sqrt{\zeta_i^2 + \alpha_i} - \zeta_i} Q + 2\zeta_i A_i^\top P A_i \succeq (A_i^\top P A_i + A_i^\top P A)^+.$$

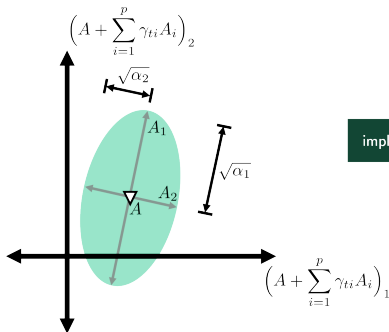
Then the deterministic system in Theorem 2 is stable for any $0 \leq \mu_i < \eta_i$, where the η_i are upper bounded by

$$\eta_i < \sqrt{\zeta_i^2 + \alpha_i} - \zeta_i < \sqrt{\alpha_i}$$

Shows how robustness margins depend on multiplicative noise variances

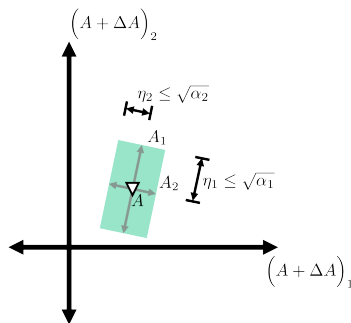
- $\sqrt{\alpha_i}$ relation
- Collapse to nothing when variances all zero
- Increase monotonically with noise variances

Mean-square stability



implies

Robust stability



Previously:

Use the *same* Lyapunov function on stochastic and uncertain deterministic systems

Now:

Use *different* Lyapunov functions on stochastic and uncertain deterministic systems

More flexible since open-loop dynamics of auxiliary stochastic system are permitted to be significantly *less stable*

Theorem 4

Suppose the stochastic system

$$x_{t+1} = \left(A \sqrt{1 + \sum_{i=1}^p \eta_i} + \sum_{i=1}^p \gamma_{ti} A_i \right) x_t$$

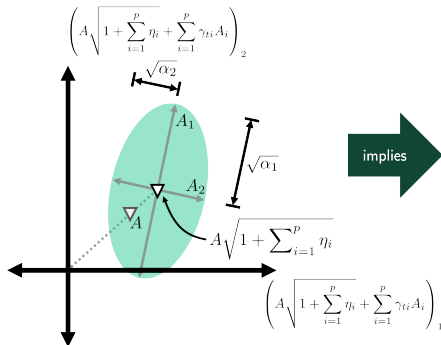
with $\mathbb{E}[\gamma_{ti}^2] = \alpha_i \geq \eta_i \left(1 + \sum_{j=1}^p \eta_j\right)$, $\eta_i \geq 0$ is mean-square stable.

Then the deterministic system

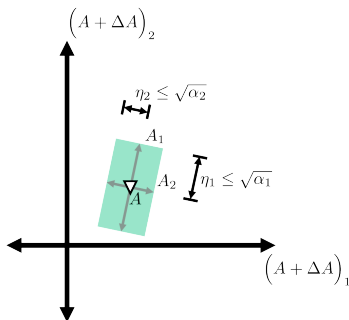
$$x_{t+1} = \left(A + \sum_{i=1}^p \mu_i A_i \right) x_t$$

is stable for all $|\mu_i| < \eta_i < \sqrt{\alpha_i}$.

Mean-square stability



Robust stability



The bounds in Theorems 2 and 4 can be optimized

Let $\theta_i \geq 0$ be scalars such that $\sum_{i=1}^p \theta_i = 1$

- These weights denote relative uncertainty in directions A_i

Largest robust stability bounds obtained as $\eta_i = y\theta_i$ and maximizing scalar y via bisection

Uncertainties in the input matrix B under closed-loop state feedback are also handled by Theorems 2 and 4 by substituting

$$\begin{aligned} A &\leftarrow A + BK, \\ \{A_i\} &\leftarrow \{A_i\} \cup \{B_j K\}, \\ \{\alpha_i\} &\leftarrow \{\alpha_i\} \cup \{\beta_j\} \end{aligned}$$

yielding a set of $p + q$ robustness bounds $\{\eta'_i\} = \{\eta_i\} \cup \{\psi_j\}$ which ensure stability of

$$x_{t+1} = \left(A + BK + \sum_{i=1}^p \mu_i A_i + \sum_{j=1}^q \nu_j B_j K \right) x_t$$

where $0 \leq \mu_i < \eta_i$, $0 \leq \nu_j < \psi_j$

Infinite-horizon LQR problem with multiplicative noise

$$\begin{aligned} \min_{u_t} \quad & \mathbb{E}_{\{\gamma_{ti}\}, \{\delta_{tj}\}} \sum_{t=0}^{\infty} (x_t^\top Q x_t + u_t^\top R u_t), \\ \text{s.t.} \quad & x_{t+1} = \left(A + \sum_{i=1}^p \gamma_{ti} A_i \right) x_t + \left(B + \sum_{j=1}^q \delta_{tj} B_j \right) u_t, \end{aligned}$$

where $Q \succeq 0$ and $R \succ 0$.

Solution

Linear state feedback $u_t = K^* x_t$ where optimal gains K^* are found by solving the generalized algebraic Riccati equation (GARE)

$$P = Q + \sum_{i=0}^p \alpha_i A_i^\top P A_i - A^\top P B \left(R + \sum_{j=0}^q \beta_j B_j^\top P B_j \right)^{-1} B^\top P A,$$

where $A = A_0$, $B = B_0$, $\alpha_0 = 1$, $\beta_0 = 1$.

Our main results lead to two algorithms for generating controllers

General flow:

1 Inputs:

- Nominal system
- Cost matrices
- Uncertainty directions and relative weights

2 Processing:

- Find largest scaling of noise variances that admits a solution to GARE

3 Outputs:

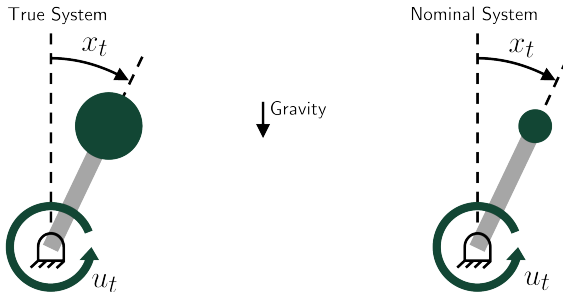
- Cost-optimal, maximally robust control law
- Robustness margins

Classic example: inverted pendulum with torque actuator

$$A = \begin{bmatrix} 1 & \Delta t \\ m_c \Delta t & 1 \end{bmatrix}, \quad B = \begin{bmatrix} 0 \\ \Delta t \end{bmatrix}$$

Take a step size $\Delta t = 0.1$.

True mass constant $\bar{m}_c = 10$, nominal model underestimates as $m_c = 5$.



Sets of true $\bar{A}$ matrices stabilized by controls from Algos. 1 and 2:

$$\bar{A} \in \begin{bmatrix} 1 & 0.1 \\ 0.1 + \mu_1 & 1 \end{bmatrix}$$

where $|\mu_1| < 3.970$ and $|\mu_1| < 6.997$ for Algos. 1 and 2 respectively.

- Robust controls guaranteed to stabilize true system
- Certainty-equivalent control failed to stabilize the true system

Our two algorithms are complementary from a control design standpoint

Code available on GitHub at

<https://github.com/TSummersLab/robust-control-multinoise>.

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